SOLUTIONS

SECTION A

1. (a) Let $k \in \mathbb{N}$. Let S_k be defined by $S_k = (1 - \frac{1}{k}, 2 + \frac{1}{k}]$.

Determine each of the following sets and express your answer as an interval or as a union of intervals:

(i) $\cup_{k=1}^{\infty} S_k$ (ii) $\cap_{k=1}^{\infty} S_k$ (iii) $\cup_{k=1}^{\infty} (S_k)^c$ (iv) $\cap_{k=1}^{\infty} (S_k)^c$.

Answer (moderate): (8)

(i) $\cup_{k=1}^{\infty} S_k = (0, 3]$ (ii) $\cap_{k=1}^{\infty} S_k = [1, 2]$

(iii) $\cup_{k=1}^{\infty} (S_k)^c = (-\infty, 1) \cup (2, \infty)$

(iv) $\cap_{k=1}^{\infty} (S_k)^c = (-\infty, 0] \cup (3, \infty)$. (8)

(b) Give a counterexample to each of the following false statements. In each case explain why the example you have given is a counterexample.

(i) The isolated points of a subset of $\mathbb{R}$ form a closed set.

(ii) Every open subset of $\mathbb{R}$ contains at least two points. (4)

Answer (moderate):

(i) The isolated points of this set $\{x | x = \frac{1}{n}, n = 1, 2, \cdots\}$ is the set itself and this is not a closed set.

(ii) The empty set $\emptyset$ is an open set and has no points. (4)

(c) Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} \dfrac{|x|\sqrt{x^2 - 4x + 4}}{x - 2} & \text{if } x \neq 2 \\ 2 & \text{if } x = 2. \end{cases}$$

Show why the function f is not differentiable at $x_0 \in \{0, 2\}$, and state the set of points where f is differentiable. Justify your answers. (10)

Answer (easy): The function f is differentiable for all $x \in \mathbb{R} \setminus \{0, 2\}$ because it is composition, product, and ratio of differentiable functions at those points. (2)

At $x = 2$ the function f is not continuous because

$$\lim_{x\to 2+} f(x) = \lim_{x\to 2+} \frac{x\sqrt{(x-2)^2}}{x-2} = \lim_{x\to 2+} \frac{x|x-2|}{x-2} = \lim_{x\to 2+} \frac{x(x-2)}{x-2} = 2$$

while

$$\lim_{x\to 2-} f(x) = \lim_{x\to 2-} \frac{x\sqrt{(x-2)^2}}{x-2} = \lim_{x\to 2-} \frac{x(-x+2)}{x-2} = -2.$$

Hence the function f is not differentiable at $x = 2$. (4)

The function f is not differentiable at $x = 0$ because

$$\lim_{x\to 0+} \frac{f(x)-f(0)}{x} = \lim_{x\to 0+} \frac{\frac{x\sqrt{(x-2)^2}}{x-2}}{x} = \lim_{x\to 0+} \frac{\sqrt{(x-2)^2}}{x-2} = -1$$

while

$$\lim_{x\to 0-} \frac{f(x)-f(0)}{x} = \lim_{x\to 0-} \frac{\frac{-x\sqrt{(x-2)^2}}{x-2}}{x} = \lim_{x\to 0-} \frac{-\sqrt{(x-2)^2}}{x-2} = 1.$$

In conclusion the function f is differentiable only on $\mathbb{R} \setminus \{0, 2\}$. (4)

(d) Suppose we have $a, b \in \mathbb{R}$ and $a < b$. Consider a function $f : [a, b] \to \mathbb{R}$. Let $g : [a, b] \to \mathbb{R}$ be defined by

$$g(x) = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0. \end{cases}$$

(i) Prove the statement: If f is continuous at $x_0 \in [a, b]$, then so is g.

(ii) Is the converse of this statement true? Justify your answer with a proof or a counterexample.

(9)

Answer (moderate): The proof is an application of the theorem about continuity of composite functions: $f(x)$ is continuous at x_0, $h(y) = |y|$ is continuous on $\mathbb{R}$ (so in particular $f(x_0)$ is an interior point of D_h), hence $g(x) = (h \circ f)(x) = |f(x)|$ is continuous at x_0.

The converse is not true, the function

$$f(x) = \begin{cases} -1 & \text{if } x \geq 0 \\ 1 & \text{if } x < 0 \end{cases}$$

is not continuous at 0, while the function $|f(x)| = 1$ is continuous on $\mathbb{R}$.

(e) For an integer k, define the set S_k by $S_k = \{k^n | n \in \mathbb{N}\}$.
Find the set K of all integers k with the property that the set S_k is bounded below. Justify your answer.

(10)

Answer (easy): For any integer $k \leq -2$, the set S_k does not have a lower bound: $\forall a \in \mathbb{R}$, we can find an odd number $n \in \mathbb{N}$ such that $|k|^n \geq -a$ and hence $k^n \leq a$. (4)

For $k = -1$, we have $S_k = \{-1, 1\}$, which has -1 as a lower bound. (3)

For every integer $k \geq 0$, we have $k^n \geq 0$ for all $n \in \mathbb{N}$, and so 0 is a lower bound of S_k. (3)

Thus, the answer is $K = \{k \in \mathbb{Z} : k \geq -1\}$.

(f) Let $h : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be the function defined by

$$h(x) = \frac{x + |x|(1+x)}{x} \sin\left(\frac{1}{x}\right).$$

(i) Use the ϵ-δ definition of limits to show that $\lim_{x \to 0-} h(x) = 0$.

(ii) Show that $\lim_{x \to 0+} h(x)$ does not exist.

(9)

Answer (easy): If $x < 0$, then $h(x) = -x \sin\left(\frac{1}{x}\right)$, so $\lim_{x \to 0-} h(x) = 0$, since

$$|h(x) - 0| = |x \sin\left(\frac{1}{x}\right)| \leq |x| < \epsilon$$

if $-\delta < x < 0$ with $\delta = \epsilon$.

If $x > 0$, then $h(x) = (2+x) \sin\left(\frac{1}{x}\right)$ takes on every value between -2 and 2 in every interval $(0, \delta)$, hence the limit from the right-hand side does not exist. (5) (4)

[**50**]

SECTION B

2. (a) Let $S = \left\{ \frac{(-1)^n}{n} \middle| n \in \mathbb{N} \right\}$.

(i) Determine the set of boundary points and the closure of the set S.

(ii) Is the set S open, closed, both or neither? Justify your answer. (10)

Answer (Difficult):

(i) The set of boundary points is $\partial S = S \cup \{0\}$ and the closure is $\overline{S} = S \cup \partial S = S \cup \{0\}$. (4)

(ii) The set is neither open nor closed. [This needs to be justified.] The set is not open since not every point of S is an interior point (in fact no point is an interior point). The set is not closed since it does not contain all of its limit points – it does not contain 0; alternatively, S^c is not open as 0 is not an interior point, hence S is not closed. (6)

(b) (i) Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions on $[-1, 1]$ defined by

$$f_n(x) = |x|^{1+\frac{1}{n}}.$$

Prove that this sequence converges uniformly to $f(x) = |x|$ on $[-1, 1]$.

(ii) Is a uniform limit of a sequence of differentiable functions necessarily differentiable? Justify your answer with a proof or a counterexample. (15)

Answer (moderate): [**25**]

(i) We can use the fact that f_n and f are even functions to look at the sup norm only on $[0, 1]$: we have

$$\|f - f_n\|_{[-1,1]} = \|f - f_n\|_{[0,1]} = \sup_{x \in [0,1]} |x - x^{1+1/n}|.$$

Differentiating $x - x^{1+1/n}$ we find there is a maximum on $[0, 1]$ at

$$\bar{x}_n = \left(\frac{n}{n+1} \right)^n$$

where

$$\bar{x}_n - \bar{x}_n^{1+1/n} = \frac{1}{n+1} \left(\frac{n}{n+1} \right)^n$$

hence

$$\sup_{x\in[0,1]} |x-x^{1+1/n}| = \frac{1}{n+1}\left(\frac{n}{n+1}\right)^n < \frac{1}{n+1} \leq \frac{1}{N+1} < \frac{\epsilon}{1+\epsilon} < \epsilon$$

for all $n \geq N$ with $N > \dfrac{1}{\epsilon}$, confirming uniform convergence. (10)

(ii) This is false. For example consider the functions f_n above. These are differentiable for all $x \in \mathbb{R}$ with derivative

$$f_n'(x) = \begin{cases} (1+\frac{1}{n})x^{\frac{1}{n}} & x > 0 \\ 0 & x = 0 \\ -(1+\frac{1}{n})(-x)^{\frac{1}{n}} & x < 0 \end{cases}$$

However, $\lim_{n\to\infty} f_n(x) = |x|$ is not differentiable at 0. (5)

3. (a) Use Riemann sums to show that the function $f : [1,2] \to \mathbb{R}$ given by $f(x) = x^2 - 2x$ is Riemann integrable, and then evaluate $\int_1^2 f(x)dx$. (12)

Answer (moderate): Consider a uniform partition of $[1,2]$ into n subintervals,

$$P_n = \left(1, 1+\frac{1}{n}, \ldots, 1+\frac{i}{n}, \ldots, 2\right),$$

each of width $\Delta x_i = \frac{1}{n}$. Since f is increasing on $[1,2]$, we obtain the lower Riemann sums

$$\begin{aligned} L(f;P_n) &= \sum_{i=1}^{n}\left(\left(1+\frac{(i-1)}{n}\right)^2 - 2 - 2\frac{(i-1)}{n}\right)\frac{1}{n} \\ &= \frac{1}{n}\sum_{i=0}^{n-1}\left(1+\frac{i^2}{n^2}+\frac{2i}{n}-2-2\frac{i}{n}\right) \\ &= \frac{1}{n}\sum_{i=0}^{n-1}\left(\frac{i^2}{n^2}-1\right) \\ &= -\frac{2}{3}-\frac{1}{2n}+\frac{1}{6n^2}. \end{aligned}$$

and the upper Riemann sums

$$\begin{aligned} U(f;P_n) &= \sum_{i=1}^{n} \left(\left(1+\frac{i}{n}\right)^2 - 2 - 2\frac{i}{n} \right) \frac{1}{n} \\ &= \frac{1}{n} \sum_{i=1}^{n} \left(\frac{i^2}{n^2} - 1 \right) \\ &= -\frac{2}{3} + \frac{1}{2n} + \frac{1}{6n^2}. \end{aligned}$$

Then

$$\underline{\int_0^1} f(x)dx = \sup_P L(f;P) \geq \sup_n L(f;P_n) = -\frac{2}{3}$$

and

$$\overline{\int_0^1} f(x)dx = \inf_P U(f;P) \leq \inf_n U(f;P_n) = -\frac{2}{3}$$

and since

$$-\frac{2}{3} \leq \underline{\int_0^1} f(x)dx \leq \overline{\int_0^1} f(x)dx \leq -\frac{2}{3}$$

we have

$$\underline{\int_0^1} f(x)dx = \overline{\int_0^1} f(x)dx = \int_0^1 f(x)dx = -\frac{2}{3}.$$

(b) Let $r \in \mathbb{R}$. Let $\{s_n\}_{n=1}^{\infty}$ be the sequence of real numbers defined by

$$s_n := \sum_{k=1}^{n} r^k.$$

(i) Use Cauchy's Convergence Criterion to prove that the sequence $\{s_n\}_{n=1}^{\infty}$ converges when $r \in (-1,1)$.

(ii) Is this sequence bounded when $r \in (-1,1)$? Justify your answer.

(iii) Describe the behaviour of the infinite series $\sum_{k=1}^{\infty} r^k$ when $r = 1$ and then when $r = -1$. In each case justify your answer.

(13)

Answer (moderate):

(i) If $|r| < 1$ and $m \geq n \geq N$, then

$$\begin{aligned} |s_m - s_n| &= |r^{n+1} + r^{n+2} + \cdots + r^m| \\ &\leq |r|^{n+1}(1 + |r| + \cdots + |r|^{m-n-1}) \\ &= |r|^{n+1}\frac{1 - |r|^{m-n}}{1 - |r|} < \frac{|r|^{n+1}}{1 - |r|}. \end{aligned}$$

If $\epsilon > 0$, choose N so that

$$\frac{|r|^{N+1}}{1 - |r|} < \epsilon.$$

Then

$$|s_m - s_n| < \epsilon \quad \text{if} \quad m \geq n \geq N.$$

Then $\{s_n\}$ is a Cauchy sequence, and hence it converges. (5)

(ii) Yes. Since $\{s_n\}$ is a Cauchy sequence, then it is bounded. Alternatively, this can be proved. Since $\{s_n\}$ is Cauchy, then $\forall \epsilon > 0$, there exists an $N \in \mathbb{N}$ such that if $m, n \geq N$, then $|s_n - s_m| < \epsilon$. For $\epsilon = 1$ we obtain that for all $n \geq N$

$$|s_n| = |s_n - s_N + s_N| \leq |s_n - s_N| + |s_N| < 1 + |s_N|$$

Hence if $M = \max\{|s_1|, \ldots, |s_{N-1}|, 1 + |s_N|\}$ then

$$|s_n| \leq M$$

for all $n \in \mathbb{N}$, therefore $\{s_n\}$ is bounded. (4)

(iii) Describe the behaviour of the infinite series $\displaystyle\sum_{k=1}^{\infty} r^k$ when $r = 1$ and then when $r = -1$. In each case justify your answer. When $r = -1$, we have $s_{2m} = 0$ and $s_{2m+1} = -1$ for $m \geq 0$; while when $r = 1$ we have $s_n = n$. In both cases the series diverges. (4)

4. Fix $k \in \mathbb{N}$. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by

$$f(\mathrm{v}) = (x_1^2 + x_2^2)^k,$$

where $\mathrm{v} = (x_1, x_2)^T \in \mathbb{R}^2$.

(a) Show that if $k = 2$ then the function f is continuous on $\mathbb{R}^2$. (5)

(b) Show that if $k = 2$ then the function f is differentiable at every point $\mathrm{v} \in \mathbb{R}^2$, and find its derivative. (10)

(c) Find all the stationary points of f when $k = 2$. (5)

(d) Is the function f differentiable at every point $\mathrm{v} \in \mathbb{R}^2$ when k is any natural number? Justify your answer. (5)

[**25**]

Answer (moderate):

(a) Let $q, r : \mathbb{R} \to \mathbb{R}$ be the continuous functions defined by

$$q(x) = x^2, \qquad r(x) = x^k, (k \in \mathbb{N}).$$

Then we can write

$$f(\mathrm{v}) = r(q(\pi_1(\mathrm{v})) + q(\pi_2(\mathrm{v}))),$$

which by applying the theorems of continuity in terms of coordinate functions and continuity of composition of continuous functions, makes it clear that f is continuous when $k = 2$. (5)

(b) We have

$$\partial_1 f(\mathrm{v}) = k(x_1^2 + x_2^2)^{k-1} 2x_1 = 2kx_1(x_1^2 + x_2^2)^{k-1}.$$

$$\partial_2 f(\mathrm{v}) = k(x_1^2 + x_2^2)^{k-1} 2x_2 = 2kx_2(x_1^2 + x_2^2)^{k-1}.$$

Since projections are continuous and sums and products of continuous functions are continuous, we have that $\partial_i f$ is continuous on $\mathbb{R}^2$ for each i. Hence f is differentiable at every point $\mathrm{v} \in \mathbb{R}^2$ when $k = 2$ with Jacobian (5)

$$\begin{aligned} Df(\mathrm{v}) &= \begin{pmatrix} \partial_1 f(\mathrm{v}) & \partial_2 f(\mathrm{v}) \end{pmatrix} \\ &= \begin{pmatrix} 2(2)x_1(x_1^2 + x_2^2)^{2-1} & 2(2)x_2(x_1^2 + x_2^2)^{2-1} \end{pmatrix} \\ &= 4(x_1^2 + x_2^2) \begin{pmatrix} x_1 & x_2 \end{pmatrix}. \end{aligned}$$

(5)

(c) Stationary points are where $Df(\mathrm{v}) = 0$, i.e., solutions to

$$4(x_1^2 + x_2^2)x_1 = 0$$
$$4(x_1^2 + x_2^2)x_2 = 0$$

which is $(0, 0)^T$. (5)

(d) Yes, similar to part (b). (5)

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